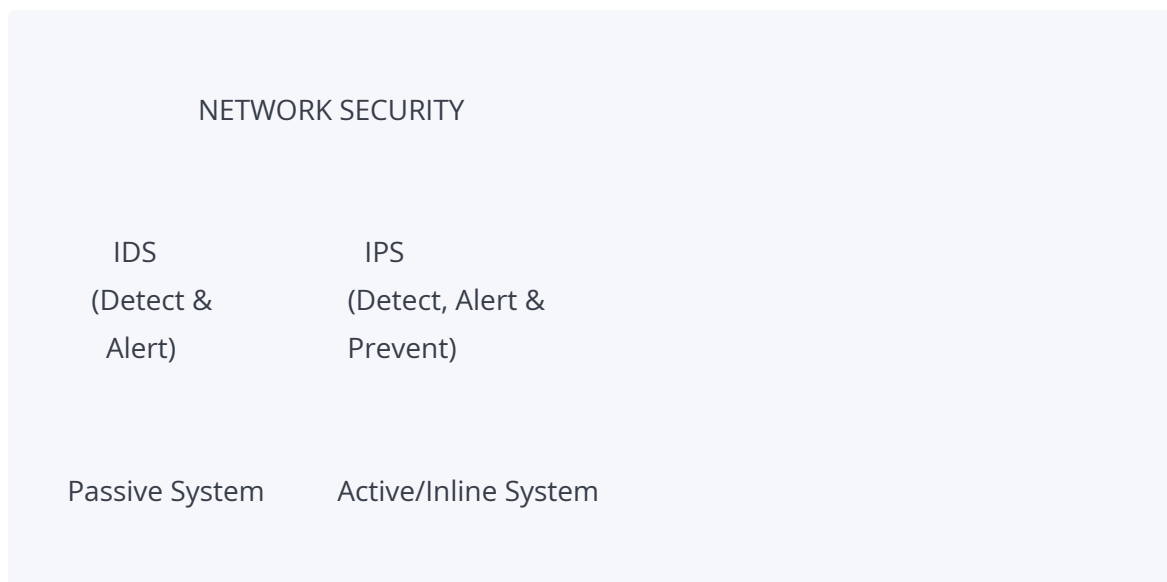


Network Defense & Countermeasures

IDS, IPS, SIEM & Network Security

SESSION : Introduction to IDS and IPS

1. IDS/IPS Overview



IDS (Intrusion Detection System)

- **Definition:** Monitors network/system activities for malicious activities or policy violations
- **Action:** Detects and **alerts** (does NOT block)
- **Mode:** Passive (out-of-band)
- **Analogy:** Security camera that records intrusion

IPS (Intrusion Prevention System)

- **Definition:** Monitors and **actively prevents** intrusions
- **Action:** Detects, alerts, and **blocks**

- **Mode:** Active (inline/in-band)
- **Analogy:** Security guard that physically stops intruder

IDS Placement:
Internet Router [IDS monitors copy] Firewall Network

IPS Placement:
Internet Router [IPS INLINE] Firewall Network

Traffic passes through IPS

2. Types of Attacks

Attack Classification Matrix

Attack Type	Description
Passive Attacks	Eavesdropping, Traffic Analysis
Active Attacks	DoS, Spoofing, Modification
Insider Attacks	Authorized user misuse
Outsider Attacks	External unauthorized access
Close-in Attacks	Physical proximity required
Distribution	Tampering hardware/software

Specific Attack Types

Attack	Description	Impact
Reconnaissance	Information gathering	Pre-attack phase
Scanning	Port/vulnerability scanning	Network mapping
Access Attacks	Unauthorized access	Data theft
DoS/DDoS	Service disruption	Availability loss
Malware	Virus, worm, trojan	System compromise

Social Engineering	Human manipulation	Credential theft
Man-in-Middle	Traffic interception	Confidentiality breach
SQL Injection	Database manipulation	Data breach

3. Security Events

Event Classification

SECURITY EVENTS

True Positive	Actual attack detected	
True Negative	Normal traffic, no alert	
False Positive	Normal flagged as attack	
False Negative	Attack NOT detected	(WORST)

Vulnerability Categories

1. Vulnerability (Weakness)

- Flaw in system that can be exploited
- Types: Software bugs, misconfigurations, weak passwords

2. Design Vulnerabilities

- Flaws in architecture/protocol design
- Example: TCP/IP protocol weaknesses (SYN flood possible)
- Example: WEP encryption design flaw

3. Implementation Vulnerabilities

- Correct design but poor implementation
- Buffer overflows, integer overflows
- Example: Heartbleed (OpenSSL implementation flaw)

Vulnerability Exploit Attack Impact

Weakness Threat Actor

SESSIONS : Advanced IDS/IPS Concepts

1. Attack Types: Traditional vs Distributed

Traditional Attacks

Single Attacker Single Target

One source IP
Easier to block
Lower bandwidth

Distributed Attacks

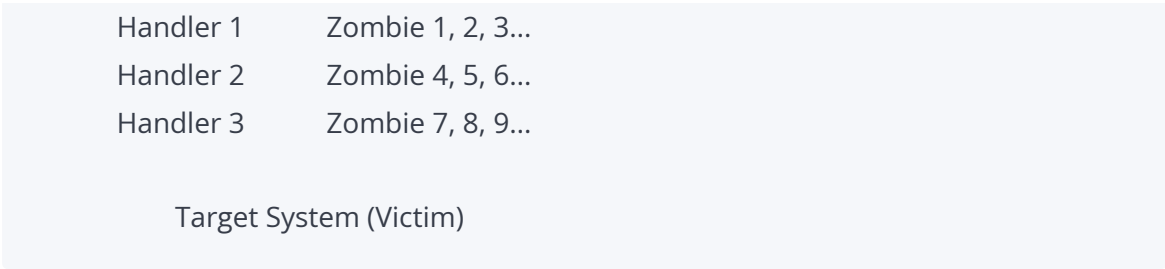
Multiple Bots (Zombies)

Single Target

- Thousands of sources
- Hard to block (IP diversity)
- Massive bandwidth
- Uses Botnet infrastructure

DDoS Architecture

Attacker (Master)

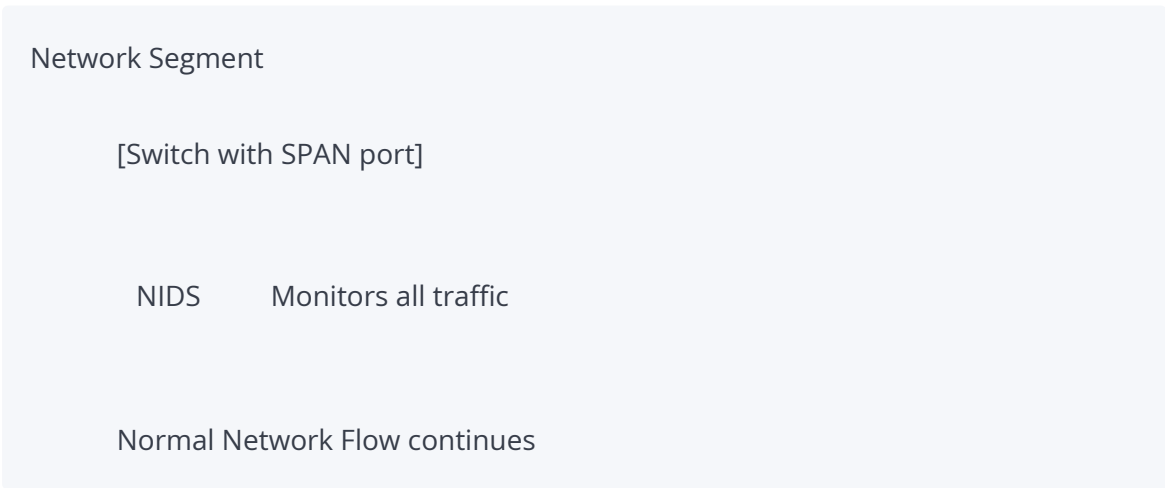


2. Intruder Types

Type	Skill Level	Motivation	Example
Script Kiddie	Low	Fun/Fame	Uses pre-built tools
Hacker	Medium-High	Curiosity/Challenge	Explores systems
Cracker	High	Financial gain	Breaks security
Insider	Varies	Revenge/Money	Disgruntled employee
State Sponsored	Very High	Espionage	APT groups
Hacktivist	Medium	Ideology	Anonymous
Cybercriminal	High	Financial	Ransomware gangs

3. Types of IDS

A. Network-Based IDS (NIDS)



- **Monitors:** Network traffic

- **Placement:** Network segment, DMZ
- **Tools:** Snort, Suricata, Zeek
- **Pros:** Single system monitors entire network
- **Cons:** Cannot decrypt encrypted traffic

B. Host-Based IDS (HIDS)

Individual Host

Server/Workstation

HIDS Agent
(OSSEC, etc.) Monitors system calls,
logs, file integrity

- **Monitors:** System logs, file integrity, processes
- **Placement:** On each host
- **Tools:** OSSEC, Tripwire, AIDE
- **Pros:** Can detect encrypted attacks, insider threats
- **Cons:** Requires installation on each host

C. Hybrid IDS

- Combines NIDS + HIDS
- Comprehensive coverage
- Centralized management

Comparison Table

Feature	NIDS	HIDS
Traffic monitoring		Limited
Log analysis	Limited	
File integrity		
Encrypted traffic		
Performance impact	Low	Medium

Cost	Lower	Higher
------	-------	--------

4. IPS Categories

A. Network-Based IPS (NIPS)

- Inline deployment
- Blocks malicious traffic in real-time
- Example: Suricata in IPS mode

B. Host-Based IPS (HIPS)

- Software on individual hosts
- Monitors and blocks malicious processes
- Example: OSSEC with active response

C. Wireless IPS (WIPS)

- Monitors wireless networks
- Detects rogue APs, wireless attacks
- Example: Cisco Adaptive Wireless IPS

D. Network Behavior Analysis (NBA)

- Analyzes traffic behavior/patterns
 - Detects anomalies in flow data
 - Uses NetFlow/IPFIX
-

5. Defense in Depth

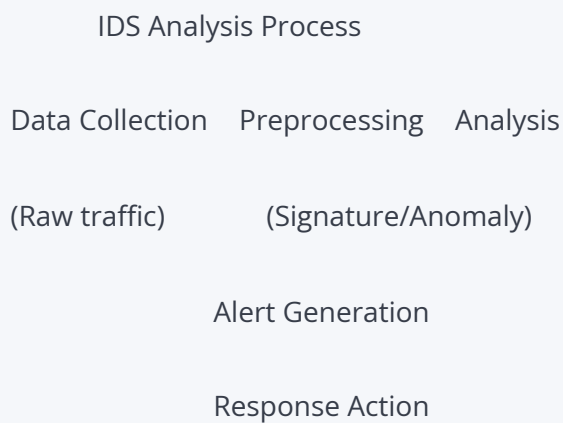
Internet

Perimeter
Security

Layer 1: Firewalls, DMZ

Network Security	Layer 2: IDS/IPS, VLAN
Host Security	Layer 3: HIDS, AV, HIPS
Application Security	Layer 4: WAF, Secure coding
Data Security	Layer 5: Encryption, DLP

6. IDS/IPS Analysis Scheme



7. Detection Methodologies

A. Signature-Based Detection

Known Attack Pattern Database		
Compare traffic	Match Found?	Alert
(Pattern matching)	(YES/NO)	

- **Pros:** Low false positives, fast detection
- **Cons:** Cannot detect zero-day attacks, requires updates
- **Example:** Virus signature databases

B. Anomaly-Based Detection

Establish Baseline (Normal behavior)	
Monitor Current Traffic	
Deviation from Baseline?	Alert

- **Pros:** Can detect unknown attacks
- **Cons:** High false positives during training
- **Example:** ML-based anomaly detection

C. Stateful Protocol Analysis

- Tracks connection states
- Compares against protocol standards
- Detects protocol violations

D. Heuristic-Based Detection

- Uses rules derived from expert knowledge
 - Behavioral analysis
 - Between signature and anomaly
-

8. Principles of IDS

1. **Accuracy:** Minimize false positives/negatives
 2. **Performance:** Real-time processing capability
 3. **Completeness:** Detect all attack types
 4. **Fault Tolerance:** Survive DoS attacks on IDS itself
 5. **Timeliness:** Quick detection and response
 6. **Scalability:** Handle growing network sizes
 7. **Adaptability:** Learn and update with new threats
-

9. Threat Hunting Model

THREAT HUNTING CYCLE

1. HYPOTHESIS
"Attacker may be using lateral movement"
 2. INVESTIGATE
Search logs, traffic, behavior
 3. DISCOVER
Identify patterns, IOCs
 4. INFORM & ENRICH
Update detection rules, share intel
 5. AUTOMATE
Create automated detection
- Back to Hypothesis

Threat Hunting Steps

1. **Trigger:** Alert or intelligence report
2. **Investigation:** Deep dive analysis
3. **Resolution:** Confirm/deny threat

SESSIONS : Attack Symptoms & Sensor Deployment

1. Symptoms of Attacks

Network Symptoms

ATTACK SYMPTOMS

Unusual Traffic Bandwidth spikes
Slow Performance High latency, packet loss
Connection Drops Unexpected disconnections
Port Scanning Sequential port connections
Unknown Processes Unfamiliar executables
Log Anomalies Deleted or modified logs
Auth Failures Multiple failed logins
Config Changes Unauthorized modifications

System Symptoms

- Increased CPU/Memory usage
- Unusual outbound connections
- New user accounts created
- Disabled security software
- Modified system files

2. Tiered Architecture (Three-Tier)

INTERNET

TIER 1: DMZ

Web Servers

Mail Servers

DNS Servers

IDS/IPS Sensors

Firewall

TIER 2: Internal

Application Layer

App Servers

Middleware

NIDS

Firewall

TIER 3: Core

Database Layer

DB Servers

File Servers

HIDS on each host

3. Sensors: Network vs Host Based

Network-Based Sensors

Network Tap/SPAN Port

Network Sensor	Captures all packets
-------------------	----------------------

Analysis Engine

Capabilities:

- Packet capture
- Protocol analysis
- Flow analysis
- Bandwidth monitoring

Host-Based Sensors

Operating System

HIDS Agent	Monitors system calls, files, logs, processes
------------	---

Central Manager

Capabilities:

- Log monitoring
- File integrity checking
- Process monitoring
- Registry monitoring (Windows)

4. DoS and DDoS

DoS Attack Types

DoS ATTACK TYPES

Volume-based	UDP flood, ICMP flood
Protocol-based	SYN flood, Ping of Death

Application layer based	HTTP flood, Slowloris
-------------------------	-----------------------

SYN Flood Attack

Attacker	Target Server
SYN (fake IP)	Server allocates resource Waits for ACK...
SYN (fake IP)	More resources allocated
(No ACK sent)	Resources EXHAUSTED Legitimate users DENIED

DDoS Mitigation Techniques

DDoS MITIGATION

- | | |
|----------------------|-----------------------|
| 1. Rate Limiting | Limit requests/sec |
| 2. IP Blacklisting | Block source IPs |
| 3. Traffic Scrubbing | Filter malicious pkts |
| 4. Anycast Diffusion | Spread traffic |
| 5. CDN | Distribute load |
| 6. SYN Cookies | Validate connections |
| 7. BGP Blackholing | Route traffic to null |

iptables Rate Limiting Commands

```
# Limit connections per IP
iptables -A INPUT -p tcp --dport 80 \
-m limit --limit 25/minute \
--limit-burst 100 -j ACCEPT

# Block after threshold
```

```
iptables -A INPUT -p tcp --dport 80 \  
-m recent --name BAD_HTTP \  
--rcheck --seconds 60 \  
--hitcount 200 -j DROP
```

SYN flood protection

```
iptables -A INPUT -p tcp --syn \  
-m limit --limit 1/s -j ACCEPT
```

5. IDS Agents

Functions of IDS Agents

IDS AGENT FUNCTIONS

1. Data Collection - Capture packets
- Read log files
2. Preprocessing - Filter noise
- Normalize data
3. Analysis - Pattern matching
- Anomaly detection
4. Alert Generation - Format alerts
- Assign severity
5. Reporting - Send to manager
- Local logging

6. IDS Manager

Functions

1. **Centralized Management:** Configure all agents
2. **Event Correlation:** Combine events from multiple agents
3. **Alert Aggregation:** Reduce duplicate alerts
4. **Reporting:** Generate security reports
5. **Response Coordination:** Coordinate countermeasures

IDS Manager

Event Correlation Engine

Database

Alert Console

Agent 1
(NIDS)

Agent 2
(HIDS)

SESSION : Log Analysis and SIEM

1. Log Analysis Introduction

What are Logs?

LOG TYPES

System Logs	OS events, errors
Application Logs	App-specific events
Security Logs	Auth, access, threats
Network Logs	Firewall, IDS, router logs

Audit Logs Compliance, user actions

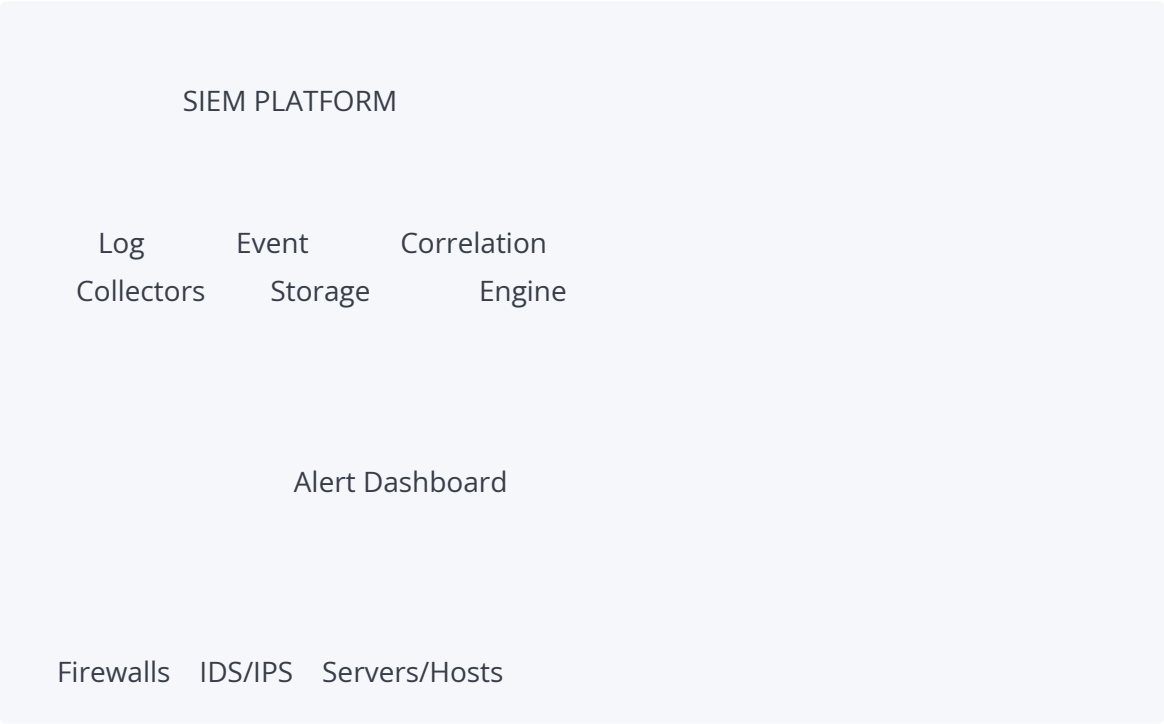
Log Components

Timestamp | Source | Event Type | Severity | Description

2024-08-15 10:23:45 | 192.168.1.50 | AUTH_FAIL | HIGH |
Failed login for user admin - attempt 5/5

2. SIEM (Security Information and Event Management)

SIEM Architecture



SIEM Key Functions

Function	Description
Log Collection	Gather logs from all sources
Normalization	Standardize log formats

Correlation	Link related events
Alerting	Notify on threats
Reporting	Compliance reports
Forensics	Historical analysis

3. SIEM Log Correlation

Correlation Rules Example

IF:

- Same source IP attempts login > 5 times in 60 seconds
- AND all attempts fail

THEN:

- Generate HIGH priority alert
- Add IP to watchlist
- Trigger incident response

Event Triggering

Raw Events Normalization Correlation Engine

Rule Matching

Alert Generated

Severity: HIGH

Category: Brute

Force Attack

4. Log Forwarding Configuration

Syslog Protocol

Log Source [UDP/TCP 514] Syslog Server
or
Log Source [TCP 6514 TLS] Secure Syslog Server

rsyslog Configuration

```
# /etc/rsyslog.conf - Client side
*. * @@192.168.1.100:514 # TCP forwarding
*. * @192.168.1.100:514 # UDP forwarding

# Server side - receive logs
$ModLoad imudp
$UDPServerRun 514
$ModLoad imtcp
$InputTCPServerRun 514
```

SESSION : Snort Testing and IDS Bypassing

1. Snort Overview

Snort Architecture

SNORT

Packet Decoder Preprocessors Detection
Engine
(Raw packets) (Normalize, (Rule matching)
defrag)

Output Plugins

(Alerts, Logs, DB)

Snort Operating Modes

1. Sniffer Mode: `snort -v`
2. Packet Logger: `snort -dev -l ./log`
3. NIDS Mode: `snort -dev -l ./log -h
192.168.1.0/24 -c snort.conf`

Snort Rule Structure

action protocol src_ip src_port direction dst_ip dst_port (options)

```
alert tcp any any -> any 80  
(msg:"HTTP Traffic"; sid:1000001; rev:1;)
```

RULE BREAKDOWN

action: alert, log, pass, drop, reject
protocol: tcp, udp, icmp, ip
src_ip: any, IP address, CIDR
src_port: any, port number, range
direction: -> (one way), <> (bidirectional)
options: msg, sid, rev, content, flags...

Sample Snort Rules

```
# Detect ICMP ping  
alert icmp any any -> any any \  
(msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

```
# Detect port scan  
alert tcp any any -> $HOME_NET any \  
(msg:"Port Scan Detected"; flags:S; \
```

```
threshold:type threshold, track by_src, \
count 20, seconds 10; sid:1000002;)
```

```
# Detect FTP login attempts
alert tcp any any -> any 21 \
(msg:"FTP Login Attempt"; \
content:"USER"; nocase; sid:1000003;)
```

```
# Detect SQL Injection
alert tcp any any -> any 80 \
(msg:"SQL Injection Attempt"; \
content:"SELECT"; nocase; \
content:"FROM"; nocase; sid:1000004;)
```

2. IDS Architecture

IDS ARCHITECTURE

Sensor Analyzer
(Collect) (Detect)

Manager
(Correlate)

Console
(Display)

3. Bypassing an IDS

Evasion Techniques

Technique	Method	Defense
Fragmentation	Split attack across fragments	Reassembly
Encoding	URL encode, Unicode	Normalization
Encryption	SSL/TLS tunneling	SSL inspection
Timing	Slow attacks (low and slow)	Behavior analysis
TTL Manipulation	Expired packets trick IDS	Protocol validation
Insertion	Insert invalid packets	Proper parsing
Obfuscation	Polymorphic shellcode	Emulation
Flooding	Overwhelm IDS with noise	Rate limiting

TTL Attack Example

Packet 1: TTL=64 Reaches both IDS and target

Packet 2: TTL=1 Reaches IDS only (IDS confused)

Packet 3: TTL=64 Completes attack on target

IDS sees: packets 1+2+3 = garbage

Target sees: packets 1+3 = complete attack

LAB CONFIGURATIONS

Lab 14: Honeypot Configuration

Honeypot Types

HONEYPOT TYPES

Low Interaction	Simulates services
	Limited risk
High Interaction	Real systems
	More data, more risk
Research	Capture new malware
Production	Detect attacks on network

Tcpdump Commands

```
# Install tcpdump
sudo apt-get install tcpdump

# Basic capture
tcpdump -i eth0

# Capture with verbosity
tcpdump -i eth0 -v

# Capture to file
tcpdump -i eth0 -w capture.pcap

# Read from file
tcpdump -r capture.pcap

# Filter by host
tcpdump host 192.168.1.100

# Filter by port
tcpdump port 80

# Filter by protocol
tcpdump tcp
tcpdump udp
tcpdump icmp

# Complex filter
tcpdump -i eth0 'tcp and port 80 and host 192.168.1.100'
```

```
# Show ASCII content
tcpdump -A -i eth0 port 80

# Show hex dump
tcpdump -X -i eth0 port 80
```

Lab 15: Snort (NIDS) vs OSSEC (HIDS)

Snort Installation (Ubuntu)

```
# Install dependencies
sudo apt-get install build-essential libpcap-dev \
    libpcap3-dev libdnet-dev zlib1g-dev

# Download and install Snort
wget https://www.snort.org/downloads/snort/snort-2.9.x.tar.gz
tar xvfz snort-2.9.x.tar.gz
cd snort-2.9.x
./configure --enable-sourcefire
make
sudo make install

# Verify installation
snort --version

# Create directories
sudo mkdir /etc/snort /etc/snort/rules /var/log/snort

# Test configuration
sudo snort -T -c /etc/snort/snort.conf
```

OSSEC Installation

```
# Download OSSEC
wget https://github.com/ossec/ossec-hids/archive/3.6.0.tar.gz
tar xvfz 3.6.0.tar.gz
cd ossec-hids-3.6.0
```



```
# Install
sudo ./install.sh

# Start OSSEC
sudo /var/ossec/bin/ossec-control start

# Check status
sudo /var/ossec/bin/ossec-control status

# View alerts
sudo tail -f /var/ossec/logs/alerts/alerts.log
```

Snort vs OSSEC Comparison

SNORT vs OSSEC

SNORT	OSSEC
NIDS	HIDS
Network traffic	System logs & files
Packet analysis	Log analysis
Signature-based	Log correlation
Real-time detect	File integrity
Low host impact	Medium host impact
Rule-based	Policy-based

Suricata IPS Setup

```
# Install Suricata
sudo apt-get install suricata

# Configure for IPS mode (NFQUEUE)
# Edit /etc/suricata/suricata.yaml

# Set mode to IPS
```

```
nfq:
  mode: accept
  fail-open: yes

# iptables rules to redirect traffic
sudo iptables -I INPUT -j NFQUEUE
sudo iptables -I OUTPUT -j NFQUEUE
sudo iptables -I FORWARD -j NFQUEUE

# Run in IPS mode
sudo suricata -c /etc/suricata/suricata.yaml -q 0

# Add custom rules
# /etc/suricata/rules/custom.rules
drop tcp any any -> $HOME_NET 22 \
(msg:"SSH Brute Force"; \
threshold:type threshold, track by_src, \
count 5, seconds 60; sid:9000001;)
```

Lab 16: ELK Stack for Threat Hunting

ELK Architecture

```
Log Sources   Filebeat/Logstash   Elasticsearch   Kibana
      (Collect)      (Store/Index) (Visualize)
```

Elasticsearch Setup

```
# Install Java
sudo apt-get install openjdk-11-jdk

# Install Elasticsearch
wget https://artifacts.elastic.co/downloads/\
elasticsearch/elasticsearch-8.x.x-amd64.deb
sudo dpkg -i elasticsearch-8.x.x-amd64.deb
sudo systemctl start elasticsearch
```

```
# Verify
curl -X GET "localhost:9200/"
```

Kibana Setup

```
# Install Kibana
wget https://artifacts.elastic.co/downloads/\
kibana/kibana-8.x.x-amd64.deb
sudo dpkg -i kibana-8.x.x-amd64.deb
sudo systemctl start kibana

# Access: http://localhost:5601
```

Logstash Configuration

```
# /etc/logstash/conf.d/snort.conf
input {
  file {
    path => "/var/log/snort/alert"
    type => "snort"
  }
}

filter {
  if [type] == "snort" {
    grok {
      match => { "message" => "%{SNORT_RULE}" }
    }
    date {
      match => ["timestamp", "MM/dd-HH:mm:ss.SSSSSS"]
    }
  }
}

output {
  elasticsearch {
    hosts => ["localhost:9200"]
    index => "snort-alerts-%{+YYYY.MM.dd}"
  }
}
```

```
}  
}
```

Lab 17: Wireshark Attack Analysis

Wireshark Filters for Attack Detection

```
# Port scan detection  
tcp.flags.syn==1 and tcp.flags.ack==0  
  
# SYN flood  
tcp.flags.syn==1 and tcp.flags.ack==0  
  and ip.src==<attacker_ip>  
  
# ARP Poisoning  
arp.duplicate-address-detected  
  
# DNS amplification  
dns and udp.length > 512  
  
# HTTP brute force  
http.request.uri contains "login"  
  and http.response.code==401  
  
# ICMP flood  
icmp.type==8 and frame.len < 100
```

Lab 18: DoS Simulation and Mitigation

LOIC (Low Orbit Ion Cannon) Analysis

LOIC Attack High volume TCP/UDP/HTTP requests
 Target CPU/Bandwidth exhaustion

Detection Indicators:

- Massive requests from single IP
- Repetitive packet patterns
- Source port randomization

iptables DDoS Mitigation

```
# Rate limiting - HTTP
iptables -A INPUT -p tcp --dport 80 \
-m hashlimit \
--hashlimit-upto 100/min \
--hashlimit-burst 200 \
--hashlimit-mode srcip \
--hashlimit-name http_limit \
-j ACCEPT

# Block after limit exceeded
iptables -A INPUT -p tcp --dport 80 -j DROP

# SYN flood protection
iptables -N SYN Flood
iptables -A SYN Flood -m limit \
--limit 5/second --limit-burst 20 -j RETURN
iptables -A SYN Flood -j DROP
iptables -A INPUT -p tcp --syn \
-j SYN Flood

# IP blacklisting
iptables -A INPUT -s 10.0.0.5 -j DROP

# Connection limit per IP
iptables -A INPUT -p tcp --syn \
-m connlimit --connlimit-above 20 -j DROP

# Save rules
sudo iptables-save > /etc/iptables/rules.v4
```

Lab 19: SIEM Configuration

Snort Rule Writing Reference

```
# Rule Header Options
alert    - generate alert and log
log      - log only
pass     - ignore packet
drop     - block and log (IPS mode)
reject   - block, log, send RST

# Detection Options
content:"string" - match string
nocase      - case insensitive
rawbytes    - match raw packet
depth:N     - search first N bytes
offset:N    - start matching at byte N
pcre:"regex" - Perl regex match
flags:SF    - TCP flags (SYN+FIN)
ttl:<5      - TTL comparison
dsize:>100  - packet size

# Threshold Options
threshold:type limit, track by_src, count 5, seconds 10;
```

BASE (Basic Analysis and Security Engine)

```
# Install BASE
sudo apt-get install php php-mysql apache2

# Configure BASE database
mysql -u root -p
CREATE DATABASE snort;
GRANT ALL ON snort.* TO snort@localhost;

# Configure BASE
# Edit base_conf.php
$DBlib_path = "adodb/";
$alert_dbname = "snort";
$alert_host = "localhost";
```

```
$alert_user = "snort";  
$alert_password = "password";
```

Lab 20: Nagios Monitoring

Nagios Installation

```
# Install Nagios  
sudo apt-get install nagios4 \  
    nagios-plugins nagios-nrpe-plugin  
  
# Start Nagios  
sudo systemctl start nagios4  
  
# Access: http://server/nagios4
```

Nagios Configuration Structure

```
/etc/nagios4/  
    nagios.cfg      (Main config)  
    cgi.cfg         (CGI settings)  
    objects/  
        contacts.cfg (Admin contacts)  
        templates.cfg (Host/service templates)  
        hosts.cfg    (Host definitions)  
        services.cfg (Service definitions)  
    conf.d/          (Custom configs)
```

Windows Nagios Agent (NSClient++)

```
; nsclient.ini  
[/settings/default]  
password = secret_password  
allowed hosts = 192.168.1.100  
  
[/settings/NRPE/server]
```

```
allow arguments = true
```

```
[/modules]
```

```
CheckSystem = enabled
```

```
CheckDisk = enabled
```

```
CheckEventLog = enabled
```

```
NSClientServer = enabled
```

```
NRPEServer = enabled
```

Linux NRPE Configuration

```
# Install NRPE on monitored host
```

```
sudo apt-get install nagios-nrpe-server \
nagios-plugins
```

```
# Configure /etc/nagios/nrpe.cfg
```

```
allowed_hosts=127.0.0.1,192.168.1.100
```

```
command[check_disk]=/usr/lib/nagios/plugins/\
```

```
check_disk -w 20% -c 10% -p /
```

```
command[check_mem]=/usr/lib/nagios/plugins/\
```

```
check_mem -w 80 -c 90
```

```
# Start NRPE
```

```
sudo systemctl start nagios-nrpe-server
```

Nagios Email Alerts

```
# contacts.cfg
```

```
define contact {
```

```
    contact_name      admin
```

```
    alias             Admin User
```

```
    email             admin@company.com
```

```
    service_notification_period 24x7
```

```
    host_notification_period   24x7
```

```
    service_notification_options w,u,c,r
```

```
    host_notification_options  d,u,r
```

```
    service_notification_commands notify-service-by-email
```



```
    host_notification_commands    notify-host-by-email
}
```

Nagios Host/Service Monitoring

```
# hosts.cfg
define host {
    host_name    web-server
    alias        Web Server
    address      192.168.1.10
    check_command check-host-alive
    max_check_attempts 3
    notification_interval 60
    notification_period 24x7
    contact_groups admins
}

# services.cfg
define service {
    host_name    web-server
    service_description HTTP
    check_command check_http
    max_check_attempts 3
    check_interval 5
    notification_interval 60
    contact_groups admins
}

define service {
    host_name    web-server
    service_description Disk Usage
    check_command check_nrpe!check_disk
    max_check_attempts 3
    check_interval 10
    contact_groups admins
}
```

QUICK REFERENCE SUMMARY

Key Commands Cheatsheet

```
# TCPDUMP
tcpdump -i eth0 -w file.pcap      # Capture
tcpdump -r file.pcap             # Read
tcpdump host 192.168.1.1         # Filter host
tcpdump port 80                  # Filter port

# SNORT
snort -T -c /etc/snort/snort.conf # Test config
snort -A console -c snort.conf    # Alert mode
snort -dev -l /var/log/snort      # Log mode

# OSSEC
ossec-control start/stop/status  # Control
ossec-logtest                    # Test rules
tail -f /var/ossec/logs/alerts.log # View alerts

# IPTABLES
iptables -L -n -v                # List rules
iptables -A INPUT -s IP -j DROP  # Block IP
iptables -F                      # Flush rules
iptables-save                    # Save rules

# SURICATA
suricata -c /etc/suricata/suricata.yaml -i eth0 # IDS
suricata-update                  # Update rules
```

Tool Purpose Matrix

Tool	Type	Purpose
Snort	NIDS/NIPS	Network intrusion detection
Suricata	NIDS/NIPS	High-performance IDS/IPS

OSSEC	HIDS	Host intrusion detection
Wireshark	Analyzer	Packet analysis
Tcpdump	Sniffer	CLI packet capture
ELK Stack	SIEM	Log analysis platform
Nagios	Monitor	System/service monitoring
LOIC	Attack tool	DoS testing
iptables	Firewall	Packet filtering
BASE	Frontend	Snort alert interface

1. Understand difference between IDS (passive) and IPS (active)
2. Know signature vs anomaly detection trade-offs
3. Memorize Snort rule structure
4. Understand true/false positive/negative concepts
5. Know SIEM log correlation principles
6. Defense in depth layers is frequently tested